IGCSE 0607 Practice Questions — E2.3 — Combined Questions & Markscheme

Algebraic fractions

Total Questions: 30 — **Total Marks:** 139 Marks

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  → **Paper 2 (Non-Calculator)** *(19 Qs)* · **Paper 4 (Calculator)** *(11 Qs)*

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1 Write as a single fraction in its simplest form.

$$\frac{2x}{5} - \frac{x-1}{3}$$

..... **[2]**

2 Write as a single fraction in its simplest form.

$$\frac{2x-1}{3}+\frac{x+4}{5}$$

..... **[2]**

3 Write as a single fraction in its simplest form.

$$\frac{3x}{4} - \frac{x-2}{5}$$

..... **[2]**

4 Write as a single fraction in its simplest form.

$$\frac{5}{x+2}+\frac{2}{3x-1}$$

..... **[3]**

5 Write as a single fraction in its simplest form.

$$\frac{x+3}{4} - \frac{2x-1}{5}$$

..... **[3]**

6 Write as a single fraction in its simplest form.

$$\frac{3x}{4} - \frac{x-2}{3}$$

..... **[3]**

7 Write as a single fraction in its simplest form.

$$\frac{2x}{3} - \frac{x-4}{5}$$

..... **[3]**

8 **(a)** Write as a single fraction in its simplest form.

$$\frac{2x}{3} + \frac{x-1}{4}$$

..... **[2]**

(b) Simplify fully.

$$\frac{15x^3y^2}{5xy^4}$$

..... **[2]**

9 **(a)** Write as a single fraction in its simplest form

$$\frac{3x}{4} - \frac{2(x-3)}{5}$$

..... **[2]**

(b) Simplify

$$\frac{6a^3b}{5c^2} \div \frac{9a^2}{10bc}$$

..... **[2]**

10 **(a)** Simplify fully.

$$\frac{3x}{4} + \frac{2x}{5}$$

..... **[2]**

(b) Simplify fully.

$$\frac{5a^2b}{12} \div \frac{10ab^2}{9}$$

..... **[2]**

11 Simplify fully.

$$\frac{x^2 - 16}{x^2 - 5x + 4}$$

. . . . . [**3**]

12 Simplify fully.

$$\frac{3x^2 - 12}{x^2 + 3x - 10}$$

..... **[4]**

13 **(a)** Write as a single fraction in its simplest form.

$$\frac{3}{x-2} - \frac{2}{x+5}$$

..... **[3]**

(b) Simplify fully.

$$\frac{x^2 - 16}{x^2 - 2x - 8}$$

..... **[3]**

14 **(a)** Simplify fully.

$$\frac{x^2 - 9}{2x^2 + 5x - 3}$$

..... **[3]**

(b) Write as a single fraction in its simplest form.

$$\frac{4}{x - 2} - \frac{3}{x + 1}$$

..... **[3]**

15 **(a)** Write as a single fraction in its simplest form

$$\frac{4}{x-3} - \frac{3}{x+2}$$

..... **[3]**

(b) Simplify

$$\frac{x^2-9}{x^2+5x+6}$$

..... **[3]**

16 Simplify fully.

$$\frac{2x^2 + 5x - 12}{4x^2 - 9}$$

..... **[4]**

17 Simplify fully.

$$\frac{3x^2 - 5x - 2}{x^2 - 4} \div \frac{3x + 1}{x^2 - x - 2}$$

..... **[6]**

18 **(a)** Simplify fully

$$\frac{2x^2 - 5x - 3}{4x^2 - 1} \div \frac{x^2 - 9}{2x^2 + 5x - 3}$$

..... **[4]**

(b) Write as a single fraction in its simplest form

$$\frac{3}{x^2 - 4} - \frac{2}{x^2 + 3x + 2}$$

..... **[3]**

19 **(a)** Simplify fully.

$$\frac{2x^2 - 5x - 3}{4x^2 - 9}$$

..... **[3]**

(b) Write as a single fraction in its simplest form.

$$\frac{3}{x-2} + \frac{4}{x+1} - \frac{2x}{x^2 - x - 2}$$

..... **[4]**

20 Simplify fully.

$$\frac{x^2 - 9}{2x^2 + 5x - 3}$$

..... **[3]**

21 Write as a single fraction in its simplest form.

$$\frac{5}{x-2} - \frac{3}{2x+1}$$

..... **[3]**

22 Simplify fully.

$$\frac{2x^2 - 9x + 9}{2x^2 - 18}$$

..... **[4]**

23 Simplify fully.

$$\frac{2x^2 - 8}{x^2 + x - 6}$$

..... **[5]**

24 **(a)** Write as a single fraction in its simplest form.

$$\frac{x+3}{2x-1} - \frac{x-2}{x+4}$$

..... **[3]**

(b) Simplify fully.

$$\frac{x^2-16}{x^2+4x}$$

..... **[2]**

25 Simplify fully.

$$\frac{x^2-25}{x^2-3x} \div \frac{x^2+6x+5}{2x-6}$$

..... **[4]**

26 **(a)** Simplify fully.

$$\frac{2x^2 - 8x}{x^2 - 16} \div \frac{6x}{3x + 12}$$

..... **[4]**

(b) Write as a single fraction in its simplest form.

$$\frac{5}{x - 3} + \frac{2x + 1}{x^2 - 9}$$

..... **[3]**

27 **(a)** Write as a single fraction in its simplest form.

$$\frac{4}{x-3} - \frac{3}{2x-1}$$

..... **[3]**

(b) Simplify fully.

$$\frac{x^2-25}{2x^2-9x-5} \div \frac{x+5}{4x+2}$$

..... **[4]**

28 **(a)** Write as a single fraction in its simplest form.

$$\frac{3}{x-2} - \frac{x+5}{x^2-4}$$

..... **[3]**

(b) Simplify fully.

$$\frac{4x^2-25}{2x^2-5x} \times \frac{6x^3}{4x^2+10x}$$

..... **[4]**

29 **(a)** Write as a single fraction in its simplest form.

$$\frac{5}{x-2}+\frac{4}{2x+3}$$

..... **[3]**

(b) Simplify fully.

$$\frac{2x^2-5x-12}{4x^2-9}\div\frac{x-4}{2x-3}$$

..... **[4]**

30 **(a)** Simplify fully.

$$\frac{2x^2 + 5x - 3}{4x^2 - 1}$$

..... **[4]**

(b) Express as a single fraction in its simplest form.

$$\frac{5}{x - 3} + \frac{x + 2}{x^2 - 4x + 3}$$

..... **[4]**

1

Correct common denominator of 15 with unsimplified numerator $3(2x) - 5(x-1)$ **[M1]**

$\frac{x+5}{15}$ **[A1]**

[Back to Question] *[Total Marks: 2]*

2

M1 for a correct common denominator with at least one correct numerator, e.g., $\frac{5(2x-1)+3(x+4)}{15}$ or $\frac{10x-5}{15} + \frac{3x+12}{15}$

$\frac{13x+7}{15}$ **[A1]**

[Back to Question] *[Total Marks: 2]*

3

$\frac{5(3x)-4(x-2)}{20}$ or $\frac{15x}{20} - \frac{4x-8}{20}$ **[M1]**

$\frac{11x+8}{20}$ **[A1]**

[Back to Question] *[Total Marks: 2]*

4

Uses common denominator $(x+2)(3x-1)$ **[B1]**

Correct unsimplified numerator $5(3x-1)+2(x+2)$ **[M1]**

$\frac{17x-1}{(x+2)(3x-1)}$ or $\frac{17x-1}{3x^2+5x-2}$ **[A1]**

[Back to Question] *[Total Marks: 3]*

5

Writes fractions over a common denominator of 20 **[M1]**

Correct unsimplified numerator: $5(x+3)-4(2x-1)$ or $5x+15-8x+4$ **[M1]**

$\frac{19-3x}{20}$ or $\frac{-3x+19}{20}$ **[A1]**

[Back to Question] *[Total Marks: 3]*

6

Common denominator of 12 and correct numerator formulation: $\frac{3(3x)-4(x-2)}{12}$ **[M1]**

Numerator expanded correctly to $9x-4x+8$ **[M1]**

$\frac{5x+8}{12}$ **[A1]**

[Back to Question] *[Total Marks: 3]*

7

$\frac{5(2x)-3(x-4)}{15}$ or $\frac{10x}{15} - \frac{3(x-4)}{15}$ **[M1]**

$10x-3x+12$ or better as numerator (allow one sign error for M1, e.g., $10x-3x-12$) **[M1]**

$\frac{7x+12}{15}$ **[A1]**

[Back to Question] *[Total Marks: 3]*

8 **(a)** Uses a common denominator of 12, e.g. $\frac{4(2x)+3(x-1)}{12}$ **[M1]**

$\frac{11x-3}{12}$ **[A1]**

(b) $3x^2$ or $\frac{1}{y^2}$ seen **[M1]**

$\frac{3x^2}{y^2}$ or $3x^2y^{-2}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

9 **(a)** Common denominator 20 with numerator $5(3x) - 4 \times 2(x-3)$ or $15x - 8x + 24$ **[M1]**

$\frac{7x+24}{20}$ **[A1]**

(b) $\frac{6a^3b}{5c^2} \times \frac{10bc}{9a^2}$ **[M1]**

$\frac{4ab^2}{3c}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

10 **(a)** $\frac{15x+8x}{20}$ **[M1]**

$\frac{23x}{20}$ **[A1]**

(b) $\frac{5a^2b}{12} \times \frac{9}{10ab^2}$ **[M1]**

$\frac{3a}{8b}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

11 $(x-4)(x+4)$ seen **[B1]**

$(x-4)(x-1)$ seen **[B1]**

$\frac{x+4}{x-1}$ **[A1]**

[Back to Question] *[Total Marks: 3]*

12 $3(x^2-4)$ or $3(x-2)(x+2)$ **[B1]**

$(x+5)(x-2)$ [B2] (or [M1] for $(x+a)(x+b)$ where $ab = -10$ or $a+b = 3$)

$\frac{3(x+2)}{x+5}$ or $\frac{3x+6}{x+5}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

13 **(a)** $\frac{3(x+5)-2(x-2)}{(x-2)(x+5)}$ **[M1]**

$3x+15-2x+4$ correctly expanded on numerator **[M1]**

$\frac{x+19}{(x-2)(x+5)}$ or $\frac{x+19}{x^2+3x-10}$ **[A1]**

(b) $(x-4)(x+4)$ seen for numerator **[M1]**

$(x-4)(x+2)$ seen for denominator **[M1]**

$\frac{x+4}{x+2}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

14 **(a)** M1 for factorising the numerator correctly: $(x-3)(x+3)$

M1 for factorising the denominator correctly: $(2x-1)(x+3)$

$\frac{x-3}{2x-1}$ **[A1]**

(b) M1 for the correct common denominator $(x-2)(x+1)$

M1 for the correct unsimplified numerator $4(x+1)-3(x-2)$

$\frac{x+10}{(x-2)(x+1)}$ or $\frac{x+10}{x^2-x-2}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

15 **(a)** Common denominator $(x-3)(x+2)$ seen **[M1]**

Numerator $4(x+2)-3(x-3)$ or $4x+8-3x+9$ **[M1]**

$\frac{x+17}{(x-3)(x+2)}$ or $\frac{x+17}{x^2-x-6}$ **[A1]**

(b) Numerator factorised as $(x-3)(x+3)$ **[B1]**

Denominator factorised as $(x+2)(x+3)$ **[B1]**

$\frac{x-3}{x+2}$ **[B1]**

[Back to Question] *[Total Marks: 6]*

16 Factorises denominator to $(2x-3)(2x+3)$ **[B1]**

Attempts to factorise numerator to $(2x+a)(x+b)$ where $ab=-12$ or $2b+a=5$ **[M1]**

Correctly factorises numerator to $(2x-3)(x+4)$ **[A1]**

$\frac{x+4}{2x+3}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

17 $(3x+1)(x-2)$ **[B2]**

If not B2, then B1 for $(3x+a)(x+b)$ where $ab=-2$ or $3b+a=-5$ **[B1]**

$(x-2)(x+2)$ **[B1]**

$(x-2)(x+1)$ **[B1]**

$\dfrac{3x^2-5x-2}{x^2-4}\times\dfrac{x^2-x-2}{3x+1}$ or equivalent correct inverted product **[M1]**

$\dfrac{(x-2)(x+1)}{x+2}$ or $\dfrac{x^2-x-2}{x+2}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

18 (a) Factorises $2x^2-5x-3=(2x+1)(x-3)$ and $4x^2-1=(2x-1)(2x+1)$ **[B1]**

Factorises $x^2-9=(x-3)(x+3)$ and $2x^2+5x-3=(2x-1)(x+3)$ **[B1]**

Inverts second fraction and multiplies: $\dfrac{(2x+1)(x-3)}{(2x-1)(2x+1)}\times\dfrac{(2x-1)(x+3)}{(x-3)(x+3)}$ **[M1]**

1 **[A1]**

(b) Factorises denominators: $(x-2)(x+2)$ and $(x+1)(x+2)$ **[M1]**

Common denominator $(x-2)(x+2)(x+1)$ with numerator $3(x+1)-2(x-2)$ **[M1]**

$\dfrac{x+7}{(x-2)(x+2)(x+1)}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

19 (a) Factorising numerator: $(2x+1)(x-3)$ **[B1]**

Factorising denominator: $(2x-3)(2x+3)$ **[B1]**

Note: no common factor cancels; the expression is already in its simplest form

Final answer: $\dfrac{2x^2-5x-3}{4x^2-9}$ or $\dfrac{(2x+1)(x-3)}{(2x-3)(2x+3)}$ **[A1]**

(b) Recognising $x^2-x-2=(x-2)(x+1)$ **[M1]**

Writing all over a common denominator of $(x-2)(x+1)$: $\dfrac{3(x+1)+4(x-2)-2x}{(x-2)(x+1)}$ **[M2]**

Simplifying numerator: $3x+3+4x-8-2x=5x-5$ **[M1]**

$\dfrac{5(x-1)}{(x-2)(x+1)}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

20 Factorising numerator to $(x-3)(x+3)$ **[B1]**

Factorising denominator to $(2x-1)(x+3)$ **[B1]**

$\dfrac{x-3}{2x-1}$ **[A1]**

[Back to Question] *[Total Marks: 3]*

21 Denominator of $(x-2)(2x+1)$ **[B1]**

Correct unsimplified numerator: $5(2x+1)-3(x-2)$ **[M1]**

$\dfrac{7x+11}{(x-2)(2x+1)}$ **[A1]**

[Back to Question] *[Total Marks: 3]*

22 Numerator factorised as $(2x-3)(x-3)$ **[B2]**

Allow [M1] for $(2x+a)(x+b)$ where $ab=9$ or $2b+a=-9$

Denominator factorised as $2(x-3)(x+3)$ or $2(x^2-9)$ or $(2x-6)(x+3)$ **[B1]**

$\dfrac{2x-3}{2(x+3)}$ or $\dfrac{2x-3}{2x+6}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

23 $2(x-2)(x+2)$ **[B2]**

If not B2, then B1 for $2(x^2-4)$ or $(2x-4)(x+2)$ **[B1]**

$(x-2)(x+3)$ **[B2]**

If not B2, then B1 for $(x+a)(x+b)$ where $ab=-6$ or $a+b=1$ **[B1]**

$\dfrac{2(x+2)}{x+3}$ or $\dfrac{2x+4}{x+3}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

24 (a) Common denominator $(2x-1)(x+4)$ seen **[M1]**

Numerator: $(x+3)(x+4)-(x-2)(2x-1)$ **[M1]**

$(x^2+7x+12)-(2x^2-5x+2)$ correctly simplified to $-x^2+12x+10$ **[A1]**

(b) Factorising numerator to $(x-4)(x+4)$ or denominator to $x(x+4)$ **[M1]**

$\dfrac{x-4}{x}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

25 Factorising x^2-25 to $(x-5)(x+5)$ and x^2-3x to $x(x-3)$ **[B1]**

Factorising x^2+6x+5 to $(x+1)(x+5)$ and $2x-6$ to $2(x-3)$ **[B1]**

Showing multiplication by the reciprocal, e.g. $\dfrac{(x-5)(x+5)}{x(x-3)}\times\dfrac{2(x-3)}{(x+1)(x+5)}$ **[M1]**

$\dfrac{2(x-5)}{x(x+1)}$ (or equivalent expanded form $\dfrac{2x-10}{x^2+x}$) **[A1]**

[Back to Question] *[Total Marks: 4]*

26 **(a)** M1 for factorising $2x^2 - 8x$ to $2x(x-4)$

M1 for factorising $x^2 - 16$ to $(x-4)(x+4)$

M1 for inverting the second fraction and factorising, i.e., multiplying by $\frac{3(x+4)}{6x}$

1 **[A1]**

(b) M1 for writing the denominator $x^2 - 9$ as $(x-3)(x+3)$

M1 for common denominator $(x-3)(x+3)$ and correct numerator $5(x+3) + 2x + 1$

$\frac{7x+16}{(x-3)(x+3)}$ or $\frac{7x+16}{x^2-9}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

27 **(a)** $\frac{4(2x-1)-3(x-3)}{(x-3)(2x-1)}$ **[M1]**

Numerator expanded correctly to $8x - 4 - 3x + 9$ **[M1]**

$\frac{5x+5}{(x-3)(2x-1)}$ or $\frac{5(x+1)}{(x-3)(2x-1)}$ **[A1]**

(b) Multiplies by reciprocal: $\times \frac{4x+2}{x+5}$ **[M1]**

Factorises $x^2 - 25$ to $(x-5)(x+5)$ or $4x+2$ to $2(2x+1)$ **[M1]**

Factorises denominator $2x^2 - 9x - 5$ to $(2x+1)(x-5)$ **[M1]**

2 **[A1]**

[Back to Question] *[Total Marks: 7]*

28 **(a)** Uses common denominator $(x-2)(x+2)$ or $x^2 - 4$ **[B1]**

Correct unsimplified numerator $3(x+2) - (x+5)$ **[M1]**

$\frac{2x+1}{(x-2)(x+2)}$ or $\frac{2x+1}{x^2-4}$ **[A1]**

(b) Factorises $4x^2 - 25$ as $(2x-5)(2x+5)$ **[B1]**

Factorises denominators as $x(2x-5)$ and $2x(2x+5)$ **[B2]**

Allow [B1] if only one denominator is factorised correctly

Correct cancellation leading to final answer $3x$ **[A1]**

[Back to Question] *[Total Marks: 7]*

29 **(a)** $\frac{5(2x+3)+4(x-2)}{(x-2)(2x+3)}$ **[M1]**

$10x + 15 + 4x - 8$ **[M1]**

$\frac{14x+7}{(x-2)(2x+3)}$ or $\frac{7(2x+1)}{(x-2)(2x+3)}$ **[A1]**

(b) $\frac{2x^2-5x-12}{4x^2-9} \times \frac{2x-3}{x-4}$ **[M1]**

$(2x+3)(x-4)$ **[B1]**

$(2x-3)(2x+3)$ **[B1]**

1 **[A1]**

[Back to Question] *[Total Marks: 7]*

30 **(a)** $(2x-1)(x+3)$ seen for numerator **[M1]**

$(2x-1)(2x+1)$ seen for denominator **[M1]**

Identifies and cancels the common factor $(2x-1)$ **[M1]**

$\frac{x+3}{2x+1}$ **[A1]**

(b) Factorises denominator: $x^2-4x+3=(x-1)(x-3)$ **[M1]**

Writes with a common denominator: $\frac{5(x-1)+(x+2)}{(x-1)(x-3)}$ **[M1]**

Simplifies numerator: $5x-5+x+2=6x-3$ **[M1]**

$\frac{3(2x-1)}{(x-1)(x-3)}$ or $\frac{6x-3}{x^2-4x+3}$ **[A1]**

[Back to Question] *[Total Marks: 8]*